Source-only model selection and external evaluation of complete spectral-sharing pipelines in motor-imagery EEG decoding

Short title: Source-only selection and complete spectral-sharing pipelines in motor-imagery EEG

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Abstract

We examined the reliability of source-only training-duration selection and the external performance of complete fixed spectral-sharing pipelines for motor-imagery EEG decoding. An exploratory development study used nine BNCI2014_001 participants, excluding both sessions of each held-out person from fitting and duration selection. Separate frozen binary pipelines were evaluated on PhysioNet, Cho2017 and Lee2019 without target fitting or adaptation. A post-outcome BNCI sensor analysis described baseline-relative mu/beta power without decoder feedback. Mean-rank selection increased development balanced accuracy from 33.72% to 42.67%; S3 and S8 contributed 87.57% of the aggregate gain. A matched-runtime fixed-duration control improved four of nine people despite a positive mean. Shared-minus-broad external differences were +0.573 percentage points on PhysioNet (95% person-bootstrap interval [−0.097,1.242]), −1.513 on Cho2017 [−2.249,−0.804] and +0.204 on Lee2019 [−0.515,0.969]. These contrasts combine unequal source-selected durations with different input views and computation; they do not isolate spectral sharing. Archived CSP outputs showed strong left-prediction dominance on both later cohorts. The BNCI analysis found negative mean signed hand laterality with participant variation; its nine-person associations do not establish a decoder mechanism. Duration choices yielded large but concentrated development gains. The frozen shared-input pipelines showed no consistent external advantage, and uncertain contrasts establish neither superiority nor equivalence. Acquisition uncertainty and offline evaluation limit deployment conclusions.

Introduction

Motor-imagery EEG decoding must distinguish imagined actions in scalp recordings whose spatial and spectral structure varies between people. A calibration-free model should transfer to a new person without fitting to that person’s labels or distributional statistics. This requirement differs from within-person evaluation and adaptation using unlabeled target recordings. BCI Competition IV dataset 2a is a small, well-documented benchmark [1,2], but its nine participants make the

cohort mean especially sensitive to a few large changes. Reviews and multi-dataset evaluation frameworks document variation between datasets and show how evaluation design shapes reported performance [3,4].

A decoder that predicts predominantly one class can have balanced accuracy near chance even when the same architecture performs well after a different training duration. This failure may be attributed to an input distribution that requires normalization or a different representation, overlooking the role of model selection. Likewise, an increase in average balanced accuracy may reflect recovery in one or two participants rather than consistent improvement across people. Source-only fitting prevents target leakage within each fit, but it does not provide independent confirmation when the research programme was developed after examining the same benchmark.

The study has two scientific aims. The first is to audit source-only duration selection: how heterogeneous validation-loss curves influence the chosen schedule, and whether an aggregate gain reflects benefit for most participants. The second is to assess complete frozen spectral-sharing pipelines in external binary transfer, retaining adverse and uncertain results alongside developmental spectral, spatial, capacity and robustness controls. The external evidence comprises an earlier PhysioNet evaluation [5,6] and separately fitted, subsequently frozen pipelines for Cho2017 [7,8] and Lee2019 [9,10]. Their montage and time-window differences require separate reporting. A downstream BNCI sensor characterization is auxiliary and receives no role in decoder fitting or selection.

The methodological contribution is an audit of selection reliability and participant-level heterogeneity, rather than a claim that rank aggregation is a new or generally superior algorithm [11]. The empirical contribution is a frozen external evaluation of complete selected pipelines, including a corrected adverse result and uncertain estimates. Saved predictions, contracts and checksums make these claims inspectable; that evidence chain is a transparency contribution, not a new decoding method. The physiological characterization supplies descriptive sensor context rather than mechanistic validation.

Related Work

Classical motor-imagery decoders use spatial and spectral structure. CSP estimates discriminative spatial projections from class covariance structure [12], and filter-bank CSP combines spatial features across frequency bands [13,14]. These methods inform the spatial and spectral controls used here. The implemented filter bank is FBCSP-inspired rather than an exact reproduction; its projections, feature scaling and classifiers are fitted on source partitions under the same target-information restriction as the neural models.

EEGNet combines temporal filtering, depthwise spatial filtering and separable convolutions in a compact architecture [15]; broader CNN work investigates learned EEG decoding and visualization [16]. Our shared-input model processes predefined mu and beta views with one configured EEGNet and averages their class logits. Its parameter count matches one branch, but computation on two views and shared BatchNorm exposure differ. These bands define experimental input views; they do not provide verified physiological source separation.

Algorithm reviews and multi-dataset benchmarks document performance variation between methods, people and datasets [3,4]. Domain-generalization evaluation must also account for model selection [11]. Here, fitting is restricted to source participants within a repeatedly used development benchmark, followed by external evaluation of frozen pipelines. Methods that fit on target labels or unlabeled target distributions have a different information budget and were not compared directly.

Recent domain-generalization studies seek invariance across source domains for unseen-subject motor-imagery decoding. EEG-DG aligns marginal and conditional distributions across multiple source domains [17]. Zheng et al. combine spectral/task knowledge distillation with cross-source correlation alignment and distance regularization [18]. These approaches pursue source-only generalization without target adaptation, although a matched comparison still requires compatible information budgets and selection procedures. Our contribution is an audit of source-only model selection, participant heterogeneity and complete frozen-pipeline transfer, with adverse and null

results retained in a traceable evidence record. We do not introduce a new domain-generalization algorithm or rank the evaluated pipelines against these methods.

EEG pretraining offers another route to transferable representations. LaBraM learns generic representations from large EEG collections [19], while EEGPT uses pretrained transformers for transferable EEG representations [20]. Fair comparison requires the pretraining data, downstream fitting procedures and computational budgets to be specified. Neither model was evaluated here, and equivalence to our target-information contract was not established; the findings concern the selected source-only pipelines.

Event-related desynchronization and synchronization describe decreases and increases in spectral power relative to a reference interval [21]. Earlier motor-imagery work investigated sensorimotor rhythm changes [22]. These concepts guide the separate characterization of measured signals, for which the baseline and estimator are specified explicitly. Scalp power patterns do not establish anatomical source localization or demonstrate that a learned decoder used those patterns.

Multibranch designs are relevant architectural precedents, but they do not define the same evaluation. Roots et al. combine three separately parameterized EEGNet-style branches [23]; their published protocol randomly splits pooled half-second windows, rather than documenting an unseen-person external test. Luo et al. use independently learned branches for overlapping frequency bands [24], primarily in within-dataset multisubject/session evaluation. Neither design is equivalent to fixed mu/beta inputs passed through one shared backbone.

Training stability and transfer are also studied outside this information contract. Smedemark-Margulies et al. examine divergence regularization in subject-disjoint rapid-serial-visual-presentation EEG [25]; this is not a motor-imagery benchmark or a direct test of the present duration-selection rules. Zhang et al. describe target-subject fine-tuning of pretrained convolutional networks [26]. The motor-imagery task of the BEETL competition provides labelled target calibration [27], unlike the present frozen no-target-fitting transfer.

Recent PLOS ONE models illustrate further protocol differences. SMMTM uses learned multiscale temporal branches and reports within-dataset leave-one-subject-out evaluation [28]. Its validation-selection details do not establish a matching frozen external budget. RMETNet incorporates source–target distribution matching and Riemannian alignment [29], so it cannot serve as a source-only comparator without separating those operations. The protocol audit in S1 Appendix explains these distinctions without inferring unreported leakage or ranking headline scores.

Sonntag et al. relate resting spectral features to participant-specific unilateral upper-limb motor-imagery classifiers in a 26-person study [30]. Their task, resting-power estimator and within-person evaluation differ from the present nine-person post-outcome, baseline-relative sensor descriptors. That literature motivates physiological description; it does not independently validate the present associations or identify a learned decoder mechanism.

Materials and Methods

Experimental Design

The study comprises exploratory development and two external-evaluation stages. In the four-class BNCI2014_001 development stage, both sessions of each target person are excluded from the corresponding fitting and duration selection. Earlier benchmark outcomes informed later candidates, so the nine-person comparisons remain exploratory. The earlier PhysioNet external evaluation applies separately fitted binary BNCI models to PhysioNet; a metadata-driven sampling-rate amendment followed partial target execution. The frozen Cho2017/Lee2019 transfer experiment uses a new binary source pipeline with a different montage, crop and reference, evaluating its frozen broad, shared-input and CSP models on Cho2017 and Lee2019 offline-training segments. These datasets are external to fitting, but participant identity nonoverlap across providers was not independently established.

The primary external estimand is the mean paired balanced-accuracy difference between the frozen shared-input and broad pipelines, with equal weight for each person and separate estimates for each cohort. The PhysioNet evaluation uses 17 shared and 18 broad epochs; the frozen Cho2017/Lee2019 evaluation uses 19 shared and 14 broad epochs, all selected on source data. Each contrast therefore evaluates complete selected pipelines rather than the isolated effect of sharing. CSP provides context, and neither external stage compares rank with mean-loss selection. The descriptive BNCI physiological analysis was defined after decoder results were known and uses BNCI signals only. Its raw-signal summaries and associations provide no feedback into the decoders and cannot establish their physiological mechanism.

Fig 1 summarizes the study sequence and source–target information boundaries.

Fig 1. Study workflow and information boundaries. Both sessions of each internal target person are excluded from the corresponding source fit; these development analyses remain exploratory. The PhysioNet and Cho2017/Lee2019 evaluations use separate binary models and preprocessing contracts. At inference, deterministic adapters and frozen source-derived transformations are applied without fitting, adaptation or BatchNorm updates. Event and class metadata establish eligibility and canonical label mapping; ground-truth labels are used for scoring. The separate BNCI-only physiological analysis reads native signals and labels after decoder outcomes are known, without feeding back into fitting or selection. Results are aggregated by person. The earlier PhysioNet evaluation's post-partial-run amendment and the external differences in selected duration are described in the Methods and comparison-fairness table.

Datasets, Populations and Task Separation

BNCI2014_001 corresponds to BCI Competition IV dataset 2a [1,2]. Nine participants completed two recording sessions, each containing six runs of 48 trials: 12 each of left-hand, right-hand, feet, and tongue imagery. Twenty-two EEG channels and three EOG channels were sampled at 250 Hz. Acquisition used a left-mastoid reference, right-mastoid ground, 0.5–100 Hz band-pass, and 50-Hz notch; the offline operations below were applied in addition to this acquisition processing. The

primary neural input contains EEG channels only. The archived metadata contain 5,184 trials, 576 per participant and 1,296 per class. All 488 expert artifact-flagged trials remain in the primary evaluation population. Including all trials differs from the original competition's artifact-free scoring; source-clean conditions exclude flagged source-training trials but retain every target trial.

The external EEG Motor Movement/Imagery Database version 1.0.0 [5] provides a separate binary task. We include all 109 subjects and only imagery runs 4, 8, and 12; T1 and T2 denote left- and right-fist imagery. Executed-movement, bilateral-task, and rest events are excluded from this contrast. The final audited inventory contains 327 EDF files verified against official checksums and 4,918 unique trials. Although the acquisition description specifies 160-Hz sampling, the file headers identify three subjects with 128-Hz recordings. Four-class and binary balanced accuracy have different chance levels and are reported separately.

Cho2017 [7,8] provides left/right-hand imagery recordings from 52 people at a native sampling rate of 512 Hz. The audited evaluation retains 10,520 imagery trials: 49 people have 200 trials and three (S7, S9, S46) have 240. The Lee2019 motor-imagery data [9,10] contain 54 people recorded at 1,000 Hz in two sessions. We retain only EEG_MI_train, the provider's offline-training runs, giving 100 trials per session and 10,800 trials in total. EEG_MI_test, other paradigms, and non-imagery events are excluded. The provider's term "train" identifies a recording segment; these external trials were used only for inference and scoring, never for parameter fitting. The three external cohorts cannot be assumed to represent 215 independently verified unique identities across providers.

Table 1 distinguishes source and external scoring populations.

Table 1. Distinct populations and computational endpoints.

Analysis	People	Classes	Unique trials	Primary tensor
Internal BNCI	9	4	5,184	22 × 750 at 250 Hz
PhysioNet-stage binary BNCI source	9	2	2,592	22 × 480 at 160 Hz

Analysis	People	Classes	Unique trials	Primary tensor
External PhysioNet	109	2	4,918	22 × 480 at 160 Hz
Cho/Lee-stage binary BNCI source	9	2	2,592	21 × 320 at 160 Hz
External Cho2017	52	2	10,520	21 × 320 at 160 Hz
External Lee2019	54	2	10,800	21 × 320 at 160 Hz

Both sessions of each BNCI outer target are held out. The PhysioNet and Cho2017/Lee2019 evaluations have separate binary source fits and preprocessing contracts, and neural inputs contain EEG channels only. Prediction rows repeat trials across models and seeds; they are not independent participants. Lee sessions are pooled within person for the primary endpoint.

Strict Zero-calibration Information Contract

Within each evaluation fit, strict zero-calibration means that no parameter, state, preprocessing coefficient, hyperparameter, duration, seed, decision threshold or model choice is fitted or selected using the current target person's recordings or labels. Neural weights and BatchNorm statistics, shallow-model coefficients, source-fitted transforms and source-selected schedules are fixed before target inference. Target data are not used to fit normalization, covariance alignment or adaptation, update BatchNorm, stop training or select seeds. This restriction applies to each fit; it does not make the repeatedly used BNCI development programme independent of earlier target outcomes.

Target EEG provides model inputs; event and class metadata determine declared eligibility and canonical label mappings. Labels are then used to score saved predictions. Cho recordings are stored in class-specific containers, so the adapter has access to native class metadata. Fixed filtering, channel selection, resampling and common-average referencing operate on target signals without fitting a target-dependent decoder. The Cho/Lee fixed Helmert basis depends only on channel count; scalers, CSP/PCA, whitening and EOG-regression coefficients, where used, are estimated on source-training data and applied unchanged. Separate physiological summaries use

raw signals and labels but provide no feedback into bands, crops, weights, thresholds, schedules or model choice.

The absence of target fitting does not establish physical acquisition calibration. The Cho2017/Lee2019 evaluation's native-numeric-as-microvolt convention does not verify the exported voltage units. Earlier within-person and cross-session calibrated models use target-person labelled calibration and remain outside the strict zero-calibration endpoints. The source-fitted EOG regression requires synchronous target EOG and thus additional sensors, although it fits no target coefficients.

Table 2 specifies the target-information budget.

Table 2. Target information and fitting boundaries.

Operation	Information read	Target fitting	Interpretation
Eligibility, fixed channel and class maps, cue indexing	Native rate/channel/event/class metadata	None	Declared metadata adapter
Filtering, resampling, crop and Cho/Lee common-average reference	Target EEG, fixed native metadata	None	Deterministic offline operations
Cho/Lee fixed Helmert coordinates	Channel count and signal	None	Analytical CAR-subspace basis
Source-fitted scaler/CSP/PCA/whitening/EOG regression	Source data to fit; target signal to apply	None	Frozen source coefficients; EOG needs extra sensors
Neural inference	Target tensors	None; weights and BatchNorm fixed	Strict zero-calibration prediction
Scoring and separate BNCI physiology characterization	Predictions/raw signals and canonical labels	None	Downstream analyses with no decoder feedback

Operation	Information read	Target fitting	Interpretation
Within-person or cross-session calibrated models	Target-person labeled training trials	Yes	Calibrated context; separate endpoint

The restriction applies to fitting and selecting the corresponding decoder. It permits deterministic computation on target signals, metadata inspection, and label use for scoring and separate physiology. These boundaries do not make the development history outcome-blind.

Offline Preprocessing and Source-only Partitions

For internal BNCI analyses, fourth-order zero-phase Butterworth filters are applied within each native recording run before epoch extraction. The half-open interval [2.5,5.5) s after the MAT trial start corresponds to [0.5,3.5) s after the cue and yields 750 samples. The primary neural pipeline applies no baseline subtraction, additional notch, or additional re-reference. MNE-loaded values in volts are multiplied by 10^6 to give network inputs in microvolts. These offline zero-phase operations use future samples, so this pipeline does not constitute a causal online decoder.

The broad mean-loss and mean-rank EEGNet conditions use 4–40 Hz input. Restricted broad input uses 8–30 Hz, and the shared representation uses fixed mu (8–13 Hz) and beta (13–30 Hz) views. These predefined bands are experimental choices; they are neither participant-specific estimates nor physiological source separation. Any fitted scaler, CSP/PCA projection, or whitening transform is estimated only on the applicable source-training partition.

Each primary four-class outer split holds out both sessions of one participant. The eight source people contribute 4,608 trials and the target contributes 576. Sorted source IDs define four consecutive, two-person inner-validation groups. Each inner fit trains on six people/3,456 trials and validates on two people/1,152 trials. Thus, source training and validation are separated by person. Final evaluation uses three fixed seeds, with scores averaged within person before aggregation across people. BatchNorm statistics remain fixed during model evaluation and are not updated by target examples.

Training-duration Selection and Diagnostic Controls

The mean-loss rule selects the earliest epoch that minimizes validation cross-entropy averaged with equal weight across the four inner folds. Candidate epochs are 1–40. The mean-rank rule ranks epochs within each fold by validation cross-entropy, assigns average ranks to ties, averages the ranks across folds, and selects the earliest minimum. This rule is invariant to strictly increasing (order-preserving) transformations of the losses within each fold, although that property does not guarantee better target prediction. The rank follow-up reuses the 36 frozen original inner trajectories and trains 27 new final models after the schedule has been frozen. Equation (1) states the selected-epoch rule.

Selected epoch = earliest argmin over e of (1/K) Σ_k rank_k[L_k(e)]. (1)

The rank candidate was motivated by earlier results on the same BNCI participants, so the internal comparison remains exploratory despite source-only selection within each fit. A four-cell diagnostic for S3 crosses raw and source-normalized input with two and 16 fixed epochs. Here, raw input means filtered, microvolt-scaled EEG without fitted source normalization. Two cells reuse the archived original conditions and two are new diagnostic conditions preserved in the archive; the comparison involves one participant and does not provide population-level replication. Source normalization estimates per-channel mean and scale from source-training samples only.

Later controls compare fixed 20-epoch training with raw-loss-selected schedules. A matched-runtime experiment retains the historical source-selected epoch counts but retrains the 27 final models using the same runtime and raw files as the fixed-duration comparator. Epochs are not reselected under the newer runtime. Source-count controls use k = 2, 4, or 6 source participants, four predefined identity subsets per count, and the k = 8 anchor; all counts use fixed 20 epochs. The 0train-versus-1test source-session contrast has equal training budgets: 2,304 trials, 20 epochs, and 720 optimizer updates per fit. Source-count contrasts and comparisons of one source session with the both-session anchor change both the number of training examples and optimizer updates,

preventing attribution to a single causal factor. Twenty epochs was a fixed development follow-up budget, not a prospectively established optimum or a recommendation for external deployment. The population raw-versus-source-normalized comparison also changes the selected duration for S3 from 2 to 16 epochs and S5 from 13 to 12. Its aggregate difference therefore cannot be attributed solely to normalization. The fixed-duration four-cell diagnostic separates normalization and duration more directly, but only for S3.

Neural Representation and Baseline Fairness

The four-class model is Braindecode 1.5.1 EEGNet with 22 input channels, 750 samples, F1 = 8, depth multiplier D = 2, F2 = 16, temporal kernel length 64 samples, and dropout 0.25. It has 2,932 trainable parameters in the archived implementation. Training uses Adam, learning rate 0.001, batch size 64, cross-entropy, zero weight decay, and no scheduler or default augmentation. The selection seed is 20260923 and the three final seeds are 20260924, 20260925, and 20260926. This is a configured EEGNet study, not an exact reproduction of the original paper's preprocessing or splits [15].

For shared mu/beta input, both band views pass through the same EEGNet in one concatenated mini-batch. Their class logits are averaged before cross-entropy during training and before softmax during inference. Network parameters and BatchNorm are shared, while dropout acts per view. The model has no learned band weights, attention mechanism, or independently trained branches. Its parameter count matches one EEGNet, but forward-pass computation and preprocessing differ. Equation (2) defines the shared-view prediction.

$$p = \mathrm{softmax}[(f_\theta(X_\mu) + f_\theta(X_\beta))/2]. \quad (2)$$

Controls include individual frequency bands; fixed source-selected durations; Welch log-bandpower with source-fitted shallow classifiers; eight source-only CSP or PCA time-series projections per band; log-Euclidean covariance features; source-only normalization; and source-clean training. The CSP/PCA neural controls both use 16 projected channels and source-fitted

standardization. Their input widths therefore match each other but differ from the 22-channel shared-logit model. Architectural ablations use two independent networks, early band stacking, four shared bands, and a broadband capacity comparator. The independent two-band model has 5,864 parameters and its comparator has 5,914, a 0.85% mismatch. The traditional filter-bank implementation is FBCSP-inspired rather than a literal reproduction of Ang et al. [13,14].

Source-only Robustness Interventions

The six completed robustness conditions are source-pooled whitening, source-balanced empirical risk minimization (ERM), source-group distributionally robust optimization (GroupDRO), channel dropout, gain perturbation, and their combination. Source whitening uses equal-person covariance aggregation, a ridge of 5% of the mean eigenvalue, and scale preservation; no target covariance or alignment is fitted. GroupDRO treats each source person as a group [31]. Persistent log weights increase by 0.05 times detached per-group cross-entropy, are normalized by log-sum-exp, and weight the group loss. Its matched comparator is balanced ERM, which preserves the batching/sampling intervention. Training-only channel dropout uses probability 0.1 with surviving-channel scale 1/0.9; gain perturbation uses U[0.8,1.2] and precedes dropout in the combined condition. Validation and target inputs are not augmented.

Whitening and augmentation conditions are compared with the previously frozen broad rank-selected baseline. The historical baseline ran on a different runtime from the robustness batch, so these differences reflect both intervention and runtime variation. GroupDRO versus balanced ERM provides a within-runtime comparison of training objectives. This distinction accompanies the numerical table and limits attribution of cross-batch differences to the named interventions.

Earlier Binary Source Freeze and PhysioNet Evaluation

The binary BNCI source population contains 2,592 left/right trials, 288 per person. Filtering and epoching occur at 250 Hz before deterministic polyphase resampling by 16/25 to 160 Hz, giving 22 × 480 samples. The broad binary model uses 8–30 Hz; the shared model uses the fixed mu/beta

views. Source-only duration selection across all nine participants uses validation groups comprising people 1–2, 3–4, 5–6, and 7–9, with equal fold weighting. The frozen training durations are 18 epochs for the broad model and 17 for the shared model. Checkpoints, channels, labels, contrast, seeds, and preprocessing are fixed before external access. The source-only CSP4 + LDA comparator uses reg = None and LDA solver = svd; it differs from the internal CSP8 OAS/shrinkage-LDA configuration. The external neural comparison evaluates complete pipelines with different source-selected durations and therefore cannot isolate parameter sharing.

A partial external run stopped at S88 because its 128-Hz header violated the original 160-Hz assumption. A versioned amendment used metadata from the remaining headers without consulting predictions or scores. S88, S92, and S100 are filtered and epoched at their native rate, then resampled by 5/4 from 384 to 480 samples. The other 106 subjects retain native 160-Hz processing. The 87 completed subject outputs from the earlier run are copied byte-for-byte, and 22 additional subjects complete the cohort. The amendment introduces no target statistics, fitted normalization, adaptation, seed selection, or class-performance-dependent processing. Because it followed partial external execution, the combined result does not represent an unmodified execution of the original freeze.

The original aggregate validation failed because it applied an inappropriate exact floating-point equality check to re-serialized CSV probabilities. A separate cross-host runtime mismatch later prompted a versioned portability amendment. Both limitations are documented. The versioned validation-only audit checked official EDF checksums and event identities and recomputed saved probability/argmax consistency, subject metrics, and statistics. It passed with zero new fits and zero new model-inference rows. It therefore validates saved outputs; it does not regenerate all external predictions from checkpoints on the later validation host.

Frozen Cho2017/Lee2019 Transfer

The frozen Cho2017/Lee2019 study is a separate binary transfer experiment using different checkpoints from the earlier PhysioNet evaluation. All nine BNCI participants supply 2,592

left/right trials. The common ordered montage comprises Fz, FC3, FC1, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P1, Pz, P2, and POz. Each trial is processed in a fixed native-rate context from −1.5 to +4.5 s relative to the cue. Common-average reference is computed across these 21 selected channels, followed by native-rate zero-phase second-order-section Butterworth filtering with prototype order four. Broad input uses 8–30 Hz; the shared views use 8–13 and 13–30 Hz. Deterministic polyphase resampling yields 160 Hz, followed by the half-open cue crop [0.5,2.5) s, equivalent to samples [320,640) of the resampled six-second context. The resulting model tensor is 21 × 320. These preprocessing and input differences preclude interpreting a comparison with the earlier PhysioNet evaluation as an effect of external provider alone.

The Cho2017/Lee2019 transfer adapter interprets native numeric values as microvolts by convention. Physical voltage calibration, the original Cho hardware reference, and hardware cue latency have not been independently verified. The first 64 of the Cho payload's 68 rows are EEG; four EMG rows are excluded. Channel mapping follows the publication's numbered montage because the payload does not supply names. Cho native labels left = 1 and right = 2 are retained. Lee native labels left = 2 and right = 1 are deterministically mapped to canonical left = 1 and right = 2. Lee MATLAB cue indices are converted using t − 1; the provider's stored segments align with Python t, leaving a disclosed one-sample (1-ms) indexing discrepancy that was not changed using outcome scores. The supplement documents these adapter assumptions and the file-level checks.

Source-only duration selection uses four groups comprising people 1–2, 3–4, 5–6, and 7–9, candidate epochs 1–40, equally weighted mean within-fold cross-entropy ranks, and the earliest minimum. Final durations are 14 broad epochs and 19 shared epochs. The selection seed is 20260923, and final seeds are 20260924–20260926. Training uses the same configured EEGNet family, optimizer, and fixed-band logit averaging described above, with the 21-channel, 320-sample transfer dimensions. A source-fitted CSP4 + LDA comparator applies a fixed 21-to-20 Helmert coordinate transform after common-average reference, followed by source-only fitting.

The external neural comparison evaluates two complete frozen pipelines with different source-selected durations and cannot isolate parameter sharing or band decomposition alone. Source training comprises eight inner neural fits, six final neural fits, and one shallow fit: 15 in total. Migration reused these artifacts without additional source or target fitting. The broad model processes one view and the shared model two views; normalization exposure and computation also differ. These differences accompany the unequal durations in the complete-pipeline contrast. Table 3 records matched aspects and remaining differences.

Table 3. Matching and remaining differences in decoder comparisons.

Comparison	Matched aspects	Remaining difference / scope
Mean-loss versus mean-rank duration selection	Nine-person all-trial LOSO, 22×750, 4–40 Hz, original 36 inner trajectories, final seeds	Schedule changes; candidate arose after BNCI outcomes; exploratory
Raw versus source-normalized input	Same all-trial targets and EEGNet family; source-only scaler fit	Selected epochs also change: S3 $2 \rightarrow 16$, S5 $13 \rightarrow 12$; not normalization alone
S3 raw/normalized ×2/16 epochs	Epochs fixed within each raw/normalized pair	One-person diagnostic
Broad versus shared, reselected durations	Same targets, dimensions, EEGNet parameter count, optimizer/seeds	Bands, two-view computation, BatchNorm exposure and schedules differ
Broad/shared, fixed per-target schedule	Same targets and frozen mean-rank per-target epochs	Duration-controlled sensitivity; views/computation still differ
CSP versus PCA neural projections	Eight source-fitted components per band; 16 projected channels and source scaling	Matched to each other, not to 22-channel shared-logit input
Independent branches versus capacity comparator	Same targets; approximate trainable capacity	5,864 vs 5,914 parameters; branch computation/normalization differ

Comparison	Matched aspects	Remaining difference / scope
GroupDRO versus balanced ERM	Same runtime and batching/sampling policy	Declared objective comparison
Whitening/augmentation versus historical mean-rank	Same nominal target population and source-only rule	Historical baseline uses another runtime
Fixed20 versus matched-runtime CE schedule	Same actual runtime, raw files and three seeds	Historical epochs replayed, not reselected in the newer runtime
Source-session 0train versus 1test	2,304 source trials, 20 epochs, 720 updates	Session composition changes; both-session anchor also changes exposure
PhysioNet broad versus shared	109 people/4,918 trials; 22×480; same source seeds and target rule	18 vs 17 epochs; metadata amendment after partial execution
Cho2017/Lee2019 broad versus shared	Identical people/trials within cohort; 21×320 adapter, neural family/seeds	14 vs 19 epochs and one-view vs two-view processing
External CSP4+LDA context	Same cohort scoring population; source-only fit	One shallow model vs three neural seeds; Cho/Lee fixed Helmert; differs from traditional-baseline CSP8/OAS

Each comparison controls the aspects listed, while retaining the stated implementation differences. The two external evaluations therefore compare complete frozen pipelines. Matching selected aspects does not isolate every causal factor, and the CSP comparisons remain descriptive.

Official raw metadata for 18 BNCI files and all declared Cho/Lee files were audited before the preprocessing specification was committed; its pre-fit publication was recorded at 2026-10-04 01:33:03 UTC. Source training was completed at 01:52:42 UTC. The separate inference freeze was committed on 2026-10-05 at 09:31:05 UTC, before external prediction, fixing channels, timing, label adapters, transformations, checkpoints, seeds, scoring units, and primary statistical procedures. Event metadata may be inspected to determine preprocessing eligibility, but external labels are not used for fitting or model selection. BatchNorm remains in evaluation mode; there is

no target alignment, normalization fit, adaptation, or post hoc seed selection. These repository-defined freezes followed external metadata inspection and do not constitute independently registered preregistration.

All 52 Cho2017 and 54 Lee2019 participants were processed under the declared adapters. The archived independent computational validator replays processing from raw data to epochs and predictions from the frozen models, checks canonical events and scoring populations, verifies links to inference and source artifacts, and recalculates aggregate statistics. Validation passed subject to the stated acquisition-calibration limitations. The supplementary reproducibility documentation identifies the exact input, parameter, checkpoint and result revisions.

BNCI-only Physiological Signal Characterization

The BNCI characterization separately describes native BNCI2014_001 recordings: 18 files, 108 motor-imagery runs, nine people and both sessions, retaining all 5,184 four-class trials and their artifact flags in the primary population. The signals retain all 22 EEG sensors at native 250 Hz, with no additional bandpass, notch, resampling, reference transform, source scaler or learned projection. Three EOG channels are excluded. EEG column order follows the provider montage; individual MAT files do not contain channel-name strings. Native one-based trial starts are converted to zero-based samples; the cue is 500 samples after the trial start. The fixed cue-relative baseline is [−1.5,−0.5) s and the task window is [0.5,2.5) s. Before power calculation, all originals are rehashed against archived source provenance, and labels, artifact flags, finite samples, montage order and complete window bounds are checked. All 18 source SHA-256 digests, 5,184 event identities and 228,096 trial-channel-band eligibility rows passed these checks; no trial was excluded by a power guard. The primary population retained all 488 flagged trials, including 246 of the 2,592 hand trials.

Welch power is defined separately within each trial’s baseline and task windows using float64 native numeric values, a periodic Hann taper, 250-sample segments, 125-sample overlap, 250-point FFT, linear detrending, one-sided density and arithmetic segment averaging. This gives 1-Hz

bins. Closed 8–13-Hz and 13–30-Hz masks are integrated with the trapezoidal rule in physical frequency; the 13-Hz boundary supplies half-bin weights to the adjacent integrals. The trial-level change is 10 log10(Ptask/Pbaseline), in dB. Nonfinite or nonpositive powers require an explicit invalid-row receipt rather than an epsilon or clipped ratio. The 1-s baseline contributes one Welch segment and the 2-s task contributes three overlapping segments. Unequal log-power estimator variance can create a positive offset even without a true power change, so these dB summaries are not claimed to be unbiased ERD estimates.

Arithmetic mean trial dB is aggregated within person, session, class, channel and band, followed by equal averaging of the two sessions. This averages log ratios rather than taking the logarithm of averaged powers. Left/right-hand C3/C4 summaries use common trial eligibility. The signed hand descriptor is L = [(C3right − C4right) + (C4left − C3left)]/2. A negative descriptor means the contralateral task/baseline change is lower than the ipsilateral change; it does not alone establish absolute contralateral ERD. C3, Cz and C4 absolute changes, separate hand terms, full scalp summaries, sessions and artifact strata are retained to show that distinction.

The physiological parameters were fixed on 6 October 2026 after the decoder outcomes were known and before the physiological calculations. The analysis is therefore descriptive and does not provide independent confirmation. Associations use existing matched-binary BNCI LOSO scores, averaged over the three fixed neural seeds within each of the same nine people; all-nine-source fits and external-cohort scores are excluded. The broad EEGNet score is the designated descriptive endpoint. Six Spearman coefficients cover the three fixed models and two fixed bands, with no p-values, tertile groups or outcome-selected bands/models. Decoder imagery covers [0.5,3.5) s, whereas the physiological power analysis covers [0.5,2.5) s. No external physiology was calculated, and no physiological output was used to fit, select or update a decoder. The parameter freeze is commit 050e01b028aaab8e3d745934b13b2d17e9bb0a7a; its public readback preceded raw-power calculation. The preprocessing freeze, raw-data receipt, execution manifest and final

independent validation link the specification and inputs to the completed tables identified in the supplementary provenance inventory.

Statistical Analysis

Balanced accuracy = (1/C) Σ_c TP_c/(TP_c + FN_c). (3)

The held-out person is the unit of statistical inference. Paired seeds are averaged within person; predefined source-identity subsets are averaged within person and source count. Shallow conditions have one prediction per trial rather than three independent neural seeds. Paired differences are expressed in balanced-accuracy percentage points (pp). Person-bootstrap percentile intervals resample people, not trials, sessions, seeds, or source subsets. External intervals describe variation across target people conditional on the frozen source models; they do not quantify source-cohort sampling uncertainty. Internal analyses remain exploratory because benchmark outcomes informed the research program and outer training sets overlap. For Lee2019, the two retained sessions are pooled before calculating each person/seed balanced accuracy. The three neural-seed scores are then averaged within person, giving an average of scores rather than an ensemble of probabilities. A single CSP score is retained per person. Cross-cohort score differences are descriptive because the PhysioNet and Cho2017/Lee2019 evaluations differ in source preprocessing and checkpoints. Equation (3) defines the balanced-accuracy scoring rule.

For each archived experiment, we report its original statistics and the interval resampling procedure. Rank-selection intervals use 20,000 person-bootstrap draws with seed 20260923. The archived shared-minus-broad representation contrast and robustness intervals use 20,000 draws with seed 20260924. Source-composition and matched-runtime intervals use 10,000 draws with seed 20260926 and exhaustive two-sided sign flips over nine participant differences; familywise exploratory p-values use Holm correction within the frozen families. The earlier PhysioNet primary contrast uses 20,000 draws with seed 20260924 and an exact two-sided binomial sign test excluding exact zero differences. Its primary hypothesis is shared-minus-broad; contextual comparisons with CSP do not replace it. A supplementary numerical audit uses 200,000 draws

with seed 20261002 and is labelled separately. An interval excluding zero need not agree with every small-sample test. For the Cho2017/Lee2019 evaluation, the predeclared family contains exactly two shared-minus-broad contrasts, one for each external cohort. Percentile person-bootstrap intervals use 20,000 draws with seed 20260924 [32]. Two-sided person-level sign-flip tests use 20,000 Monte Carlo draws with seed 20261003, assuming sign exchangeability of paired person effects under the null, evaluating the absolute mean, and applying the (exceedances + 1)/(20,000 + 1) correction [33]. Holm adjustment controls this two-contrast family [34]; it does not retroactively include the earlier PhysioNet or internal hypotheses. The smallest attainable Monte Carlo p-value is 1/20,001. The intervals are unadjusted 95% intervals, and CSP comparisons are descriptive context. Descriptive Cho2017/Lee2019 transfer counts of positive, negative, and tied person effects use an absolute 10^−15 tolerance for ties; the frozen primary bootstrap and sign-flip calculations retain the saved numerical effects.

One historical rank-selection script labelled a nine-person binomial calculation as a sign test while retaining two unchanged participants in its denominator. We report its $p = 0.1797$ as an archived nine-person sign sensitivity. A conventional tie-excluding calculation uses seven nonzero changes and gives $p = 0.015625$; the paired t-test gives $p = 0.111$. These procedures are reported separately rather than treated as interchangeable or used to select a favourable result. Multiplicity corrections from later declared families are not retrospectively applied to unrelated earlier statistics.

The physiological characterization was specified after decoder outcomes were known and remains descriptive. Physiological aggregation and decoder associations use nine person-level records; channels, trials, sessions and seeds are not treated as independent people. The six Spearman coefficients have no physiological p-values or confirmatory testing family. They are separate from the Cho2017/Lee2019 evaluation’s frozen two-cohort tests and cannot retrospectively confirm the decoder contrasts.

Archived-output CSP diagnostic audit

The diagnostic audit reads the existing frozen CSP prediction tables, probability columns, participant identities and validation hash bindings. It reconstructs participant and pooled confusion matrices without loading a model or producing predictions. Left-prediction proportions and probability medians pool trials; the primary balanced-accuracy estimand remains an equal-person mean. A binary predicted-label complement is an arithmetic check, not a revised predictor or an outcome-selected correction. These post hoc diagnostics do not change primary scores, mappings, thresholds, models or inference contracts. They cannot by themselves exclude every adapter error or identify a physical transfer mechanism.

Reproducibility and Validation Scope

Saved predictions were independently reaggregated to check trial identities, scoring populations, person-level confusion matrices, seed-averaged scores and reported statistics. This numerical reanalysis neither loaded decoder checkpoints nor generated predictions. The Cho2017/Lee2019 experiment also underwent a completed replay from raw data to epochs and frozen-model predictions; the earlier PhysioNet portability audit checked saved outputs without replaying inference. Validation of earlier experiments remains experiment-specific, as documented in the supplement.

The physiological analysis was checked independently against native BNCI recordings. An explicit detrending, periodic-Hann FFT and trapezoidal integration calculation reproduced all 31,104 C3/Cz/C4 trial-band pairs: the maximum relative power difference was 1.85×10^{-15} and the maximum absolute dB difference was 7.11×10^{-15}, within fixed 10^{-10} tolerances. Raw spectral replay covered these three central channels; identities and eligibility were checked for all 228,096 saved rows, and all aggregation and association tables were recomputed. Numerical agreement establishes implementation consistency, not independent laboratory replication.

OpenAI Codex assisted analysis-code development and verification-code preparation. The computations use explicitly specified algorithms; the archived validation reports document the numerical checks. AI-assisted code generation is distinct from a learned decoder or a physiological model.

Ethics statement

This study involved secondary analysis of public EEG recordings and recruited no new participants. The Cho2017 and Lee2019 source publications report collection ethics review and written informed consent [7,9]. Those collection approvals are not approvals for this secondary analysis. The author confirms that neither ethics approval nor an exemption was required for this secondary analysis. No formal approval or exemption documentation was obtained for this secondary analysis; this statement records the author's assessment rather than an institutional or committee determination.

The retained research records document access to BNCI2014_001 recordings from 21 September to 6 October 2026, to PhysioNet EEG Motor Movement/Imagery recordings on 25 September 2026, and raw-data auditing and external evaluation of Cho2017 and Lee2019 MI from 3 to 5 October 2026. Dates refer to the recorded research-access periods in UTC; the calendar dates are the same in China Standard Time. They do not establish first-ever download dates or exclude access outside the retained records. Git commit, publication and backup timestamps were not substituted for recorded data-use events.

The author reports access only to the recordings, provider-assigned participant codes and basic demographic metadata, such as age, without names, contact details or an identity linkage key. No direct identifying information was available to the author; public participant codes were retained for within-dataset grouping and scoring.

Data and code availability

Public BNCI2014_001, PhysioNet, Cho2017 and Lee2019 recordings are available from the original providers [1,2,5,6,7,8,9,10], subject to their licenses and terms of use. PhysioNet platform publications are cited separately [35,36]. Analysis code, saved predictions, participant-level summaries, parameter specifications and numerical validation reports are available at https://github.com/jackzhu119/cross-subject-mi-eeg. The supplementary reproducibility documentation identifies the exact source and result revisions. Raw EEG is not redistributed in the manuscript bundle.

The preserved submitted materials and licensing notices are available at https://github.com/jackzhu119/cross-subject-mi-eeg/tree/plos-one-submission-v1.1. This post-submission revision candidate is maintained separately on paper/plos-one-postsubmission-review-20261010. Author-owned original software is licensed under MIT; author-owned original manuscript material and derived outputs are licensed under CC BY 4.0. Third-party recordings, software, fonts and incorporated materials retain their original terms. Raw EEG is not redistributed.

AI assistance and verification

ChatGPT (GPT-6, as reported by the author) assisted information retrieval, grammar revision, and preparation of instructions for publishing research materials to GitHub. OpenAI Codex assisted source and literature checks, analysis-code and verification-code development, saved-result consistency checks, figure preparation, and manuscript drafting and language revision. The author personally reviewed and revised the final manuscript and takes responsibility for its scientific content.

The underlying Codex model version is not reliably documented in the project records. Manuscript-preparation checks compare saved outputs and archived independent validation reports. The archived-output diagnostic audit involved no model fitting, checkpoint inference or

new physiological experiment. The author reports personal verification and revision of the scientific content. No AI tool is an author.

Results

Traditional baselines and heterogeneous duration selection

The all-trial four-class broad CSP8 + LDA baseline achieved 39.04% balanced accuracy; broad CSP + SVM and the full filter-bank LDA comparator achieved 37.50% and 38.12%. EEGNet with duration selected by raw mean loss achieved 33.72%, below the broad CSP baseline. These descriptive comparisons do not establish superiority of a model class. The mean-loss schedule selected only one or two epochs for S2, S3, and S8, while source-validation optima varied substantially across folds.

Source-only per-channel normalization increased mean balanced accuracy to 37.93%, a +4.21 pp difference. The archived paired 95% interval spans −1.12 to +13.49 pp, whereas the median change was only +0.23 pp. Effects varied across participants, including negative transfer in S8, so the average gain does not establish a general normalization benefit. Selected durations also changed for S3 (2 to 16 epochs) and S5 (13 to 12); the population contrast therefore combines normalization and duration changes.

The S3 diagnostic separates normalization from training duration. With two epochs, raw input gave 26.10% balanced accuracy and source-normalized input gave 25.69%. With 16 epochs, raw and normalized inputs gave 67.71% and 65.86%. The short-run raw condition assigned approximately 94% of predictions to one class. Extending training reproduced the recovery for this participant, but this four-cell diagnostic does not establish that duration explains all cross-person failures.

Mean-rank selection increased equal-person mean balanced accuracy to 42.67%, a +8.95 pp change over raw mean-loss selection. The archived bootstrap interval is [+1.05,+19.96] pp, and the median change is +1.85 pp. Seven participants improved and two were unchanged. S3 and S8

account for 87.57% of the summed improvement. The average gain therefore reflects a concentrated benefit; prior development on the same dataset and disagreement among small-sample statistics limit its generalizability.

Table 4 summarizes completed development conditions.

Table 4. Selected completed four-class conditions; all nine participants.

Condition	Bands (Hz)	Mean BA (%)	Interpretation
Broad CSP8 + LDA	4–40	39.04	Single source fit/fold
Filter-bank CSP + LDA	4–40 bank	38.12	Fixed filter-bank control
EEGNet, mean loss	4–40	33.72	Source-selected duration
EEGNet, source norm	4–40	37.93	Source-only fitted scale
EEGNet, mean rank	4–40	42.67	Reused source curves
Shared mu/beta	8–13 / 13–30	42.19	Representation-specific rank
Broad, fixed 20	4–40	43.99	Later runtime
Broad, matched CE schedule	4–40	33.92	Same runtime as fixed 20

The primary population includes all trials. Neural scores are averaged over three fixed seeds within person before population summaries. Rows span different experimental batches, so runtime variation is controlled only in explicitly matched comparisons. The table summarizes declared controls rather than a target-selected ranking; full representation and robustness inventories are in the supplement.

Fig 2 shows source-selected schedules, participant scores and the one-person duration diagnostic.

Fig 2. Training-duration selection and participant-level internal results. A: archived source-selected epochs. B: each participant’s all-trial balanced accuracy averaged over three fixed seeds; the dotted line marks four-class chance (25%). C: the S3 four-cell diagnostic, comparing raw and

source-normalized input at two fixed durations. Error bars show standard deviations across three seeds, not uncertainty across people. This diagnostic concerns one person.

Matched-runtime duration and source-composition controls

In the additive matched-runtime experiment, the historical mean-loss duration schedule gave 33.92% balanced accuracy, compared with 43.99% for fixed 20 epochs. The fixed-minus-historical-schedule mean was +10.06 pp, with a bootstrap 95% interval of [+0.68,+21.53] pp and exact sign-flip $p = 0.125$. Four of nine people improved and five worsened; the median difference was −0.174 pp. The positive mean thus reflects large recoveries rather than a typical-person benefit. The average schedule difference persisted under matched runtime, but it does not establish 20 epochs as universally preferable: historical epoch counts were reused rather than newly selected under that runtime.

Mean balanced accuracy increased descriptively with source count. Broad fixed-duration models gave 32.96%, 38.89%, 43.16%, and 43.99% at $k = 2$, 4, 6, and 8; shared models gave 34.74%, 38.89%, 42.48%, and 43.98%. Performance varied substantially with source identity at each count. The two-source broad-minus-eight-source difference was −11.03 pp, but its Holm-adjusted exploratory p was 0.164. No source-count/selection/session family showed a corrected significant result. Contrasts between the two sessions were small and heterogeneous. These trajectories are consistent with sensitivity to source training information, but do not separate source diversity from sample count or optimizer-update budget.

Fig 3 displays source-count trajectories and matched-runtime participant differences.

Fig 3. Source composition and matched-runtime controls. A: thick lines show equal-person means; thin lines show individual target-person trajectories averaged over the four predefined identity subsets and three seeds at $k = 2$, 4, 6. Every source count uses 20 epochs; $k = 8$ reuses the fixed-20 anchor. B: all nine fixed-20-minus-historical-CE-schedule effects in the matched runtime. The interval reverses the sign of the archived matched contrast. Source count also changes data volume and optimizer updates, so these exploratory sensitivities do not isolate source count alone.

Spectral, spatial, and architectural interventions

The internal shared mu/beta condition achieved 42.19% balanced accuracy, compared with 42.67% for the broad mean-rank baseline, a −0.48 pp mean difference. The archived exploratory paired interval is [−3.21,+1.41] pp (20,000 draws, seed 20260924), with exact sign-flip p = 0.914. This comparison did not demonstrate a spectral-sharing gain. The restricted broad band and individual bands provide descriptive frequency controls, while fixed-duration representation controls retain their declared schedules.

The source-only CSP8 and PCA8 neural projections achieved 41.48% and 42.06% balanced accuracy. Architectural controls achieved 42.99% for independent two-band networks, 42.19% for the broadband capacity comparator, 42.66% for early stacking, and 39.80% for four-band sharing. These alternatives provide no evidence of a general shared-band advantage. Differences in preprocessing, projected width, BatchNorm behaviour, and compute also preclude attributing the contrasts to a single physiological factor. Full model means and participant-level results accompany the manuscript.

Source-only robustness has no corrected established gain

Source-pooled whitening, source-balanced ERM, and gain perturbation had positive point estimates relative to the historical broad rank baseline, whereas dropout conditions were lower. No intervention established an improvement after correction. The matched GroupDRO-minus-balanced-ERM difference was −6.45 pp (unadjusted bootstrap interval [−11.47,−1.79] pp; exact sign-flip p = 0.046875, Holm-adjusted p = 0.140625). This adverse estimate warrants follow-up but does not establish that distributionally robust optimization generally harms EEG decoding. The whitening and gain perturbation intervals include zero, and comparison with a historical model from a different runtime further limits attribution.

Table 5 retains adverse and null robustness results.

Table 5. Exploratory source-only robustness (nine participants).

Condition	BA (%)	Control	Δ (pp)	95% CI (pp)	Holm p
Whitening	44.34	Historical mean-rank	+1.67	[−0.39, +3.67]	0.3594
Balanced ERM	43.18	Historical mean-rank	+0.51	[−0.71, +1.77]	0.4766
GroupDRO	36.73	Balanced ERM	−6.45	[−11.47, −1.79]	0.1406
Channel dropout	41.00	Historical mean-rank	−1.67	[−4.51, +0.64]	0.7617
Gain perturbation	43.42	Historical mean-rank	+0.75	[−0.34, +1.86]	0.7617
Dropout + gain	40.92	Historical mean-rank	−1.75	[−5.05, +0.91]	0.7617

Intervals are the archived, unadjusted person-bootstrap intervals; p-values use the declared within-family Holm correction. Every comparison with mean-rank follow-up crosses runtimes. GroupDRO versus balanced ERM matches batching within the same runtime. Neither intervals nor p-values are used to select a model on target outcomes.

Earlier PhysioNet transfer: a small, uncertain primary effect

In binary BNCI development, broad, shared, and CSP models achieved 68.21%, 67.35%, and 61.54% balanced accuracy. These development outcomes cannot be pooled with the four-class analyses. The separately frozen all-nine-source models then generated predictions for all 109 external subjects. Final validation confirmed 4,918 trial identities and 34,426 saved prediction rows. No model was fitted to target data.

External equal-person balanced accuracy was 61.81% for broad EEGNet, 62.39% for shared mu/beta, and 54.46% for CSP4 + LDA. The predeclared shared-minus-broad primary difference was +0.573 pp, with a 95% bootstrap interval of [-0.097,+1.242] pp. Sixty-three people favoured

shared, 43 favoured broad, and three tied; the exact tie-excluding sign-test p was 0.06446. The interval includes zero, and the sign test does not establish external superiority.

Individual performance varied widely. The shared model ranged from 39.33% to 97.07% balanced accuracy, with a median of 58.73%. Differences averaged across three seeds also varied in direction across people. The three resampled subjects had mean shared-minus-broad $\Delta = +0.514$ pp, compared with +0.575 pp for the 106 native-160-Hz subjects. The three-person subgroup comparison is descriptive and has no powered subgroup test. The result incorporates the sampling-rate and validation-portability amendments and therefore does not represent an unmodified execution of the original external protocol.

Table 6 reports the separate PhysioNet model estimates.

Table 6. Separate binary external evaluation (109 participants).

Frozen model	Mean BA (%)	Subject SD (pp)	Median BA (%)
Broad EEGNet	61.81	12.56	58.23
Shared mu/beta	62.39	12.92	58.73
CSP4 + LDA	54.46	7.90	52.08

Deep-model scores are averaged over seeds within person. The primary shared-minus-broad contrast is +0.573 pp, 95% CI [−0.097,+1.242], p = 0.06446. CSP provides context and is outside the primary hypothesis. The Methods describe the sampling-rate amendment and versioned portability validation.

Completed Cho2017 and Lee2019 transfer: lower balanced accuracy on Cho2017 and an uncertain Lee2019 contrast

The completed Cho2017/Lee2019 outputs cover 21,320 unique external trials and 149,240 saved model/seed prediction rows, seven per trial. Independent checks verified canonical labels and event identities, person-level confusion matrices, aggregate balanced accuracies, frozen statistical outputs, and links to the declared artifacts. Original source fits remained 15, with zero new

679 migration fits and zero target fits. These counts confirm source-only fitting; they do not constitute
680 independent repetitions of the external experiment.

681 On Cho2017, equal-person balanced accuracy was 59.82% for broad EEGNet and 58.31% for
682 shared mu/beta EEGNet. The primary shared-minus-broad mean was −1.513 pp (95% person-
683 bootstrap interval [−2.249,−0.804] pp), with two-sided Monte Carlo sign-flip p = 0.000049998 and
684 two-cohort Holm-adjusted p = 0.000099995. The raw p-value is the Monte Carlo procedure's
685 minimum attainable value and does not resolve probabilities below that value. Twelve people
686 favoured shared, 37 favoured broad, and three were tied within a 10^−15 numerical tolerance.
687 Shared performed worse in this declared complete-pipeline comparison after correction.

688 On Lee2019, broad and shared models achieved 65.51% and 65.72% balanced accuracy. The
689 primary difference was +0.204 pp (95% interval [−0.515,+0.969] pp); raw and Holm-adjusted p-
690 values were both 0.60367. Twenty-three people favoured shared, 30 favoured broad, and one tied.
691 This small, uncertain contrast establishes neither a reliable improvement nor equivalence. The two
692 sessions were combined within each person, so neither the 108 session records nor repeated model
693 seeds increase the primary sample size beyond 54.

694 Table 7 shows cohort scores and class recalls.

695 Table 7. Frozen external binary evaluation on Cho2017 and Lee2019.

Cohort / model	BA (%)	Person SD (pp)	Left recall (%)	Right recall (%)
Cho2017 / Broad EEGNet	59.82	8.64	46.79	72.86
Cho2017 / Shared mu/beta	58.31	7.22	37.52	79.10
Cho2017 / CSP4 + LDA	51.99	5.67	94.10	9.88
Lee2019 / Broad EEGNet	65.51	10.97	75.56	55.46

Cohort / model	BA (%)	Person SD (pp)	Left recall (%)	Right recall (%)
Lee2019 / Shared mu/beta	65.72	10.76	72.69	58.75
Lee2019 / CSP4 + LDA	52.77	5.57	85.98	19.56

Neural scores and class recalls are averaged over three seeds within person, then equally over people. Person SD describes heterogeneity, not a standard error. Lee includes only offline-training runs, pooled across two sessions. The primary shared-minus-broad contrasts are −1.513 pp [−2.249,−0.804], Holm p = 0.000100 for Cho, and +0.204 pp [−0.515,+0.969], Holm p = 0.604 for Lee. Class recalls and CSP comparisons are descriptive; no additional confirmatory tests were added.

The CSP comparator was near binary chance on both cohorts and strongly favoured left-hand labels. On Cho2017, mean left-hand recall fell from 46.79% with broad input to 37.52% with shared input, while right-hand recall increased from 72.86% to 79.10%. On Lee2019, the corresponding recalls changed from 75.56% to 72.69% and from 55.46% to 58.75%. Balanced accuracy summarizes these asymmetric shifts but does not show which class gains or loses recall. These descriptive checks used the frozen outputs and did not trigger any threshold, label, adapter, or model change. A post-submission audit of the unchanged prediction files found trial-pooled left-prediction proportions of 92.07% for Cho2017 and 83.21% for Lee2019. Median stored left-class probabilities were 0.669 and 0.633. These probability summaries do not assess calibration. Cho participants have unequal trial counts, so pooled prediction proportions must not be confused with the primary equal-person balanced accuracy. Participant confusion matrices and aggregation definitions are provided in S1 Appendix, Table S5 and S1 Data.

Fig 5 presents the frozen Cho2017/Lee2019 complete-pipeline contrasts.

Fig 5. Frozen Cho2017/Lee2019 transfer external evaluation. Cohorts are shown separately: person/model scores, equal-person means and person standard deviations, and frozen paired

person-bootstrap 95% intervals. Deep-model scores average three seeds within person; Lee sessions are pooled before person scoring. Holm p-values refer to the predeclared two-cohort shared-minus-broad family. The common 21-channel, two-second pipeline differs from the earlier PhysioNet evaluation, and no pooled cross-cohort test was specified.

Descriptive BNCI Sensor Physiology and Decoder Associations

All 18 raw files, 108 labelled runs and 5,184 trials passed the fixed coverage checks. All 228,096 trial-channel-band rows had finite, positive baseline and task power. No trial, session or person was excluded; the 488 provider-flagged trials remained in the primary summaries. Fig 6 shows equal-person scalp means and all nine C3/Cz/C4 profiles. Task/baseline changes calculated with the fixed estimator varied by sensor, imagined hand and person.

Fig 6. Descriptive native BNCI task/baseline power changes. A–D: equal-person hand-imagery means for mu (8–13 Hz) and beta (13–30 Hz) at all 22 EEG sensors. In the schematic standard-1020 geometry, anterior is upward and left scalp is on the viewer's left; linear interpolation is restricted to the sensor convex hull. No sensor or interpolated grid value is clipped by the fixed ±6-dB range. E–F: all nine equal-session C3/Cz/C4 profiles (thin lines) and equal-person means (thick lines). Trial dB uses cue baseline [−1.5,−0.5) s and task [0.5,2.5) s, with one versus three overlapping Welch segments. Negative values indicate lower task power under this estimator; unequal window precision can bias log ratios. All trials and artifact flags are retained. Template topography neither localizes cortical sources nor attributes decoder decisions. The fixed ±6-dB scale makes the small magnitude of the cohort means visible without rescaling each panel. Circle/triangle symbols distinguish imagined hands in the profiles; signed values and the shared labelled scale supplement colour. This schematic sensor map is not a cortical source map.

Mu signed laterality averaged −0.250 dB, with six negative and three positive person descriptors; beta averaged −0.168 dB, with eight negative and one positive descriptor (Table 8). Contralateral and ipsilateral group means were both negative in each band, but absolute contralateral changes were not negative for every person. For example, S2 had negative mu laterality (−0.203 dB) even

though both its contralateral (+1.491 dB) and ipsilateral (+1.694 dB) means were positive. Separate session, hand and flagged/unflagged summaries are available in the verified supplementary tables as descriptive sensitivity checks; they do not replace the all-trial population.

Table 8. Baseline-relative hand power and signed C3/C4 laterality.

Band	Contra. mean (dB)	Ipsi. mean (dB)	Mean L (dB)	Person L range (dB)	L signs (−/+)
Mu, 8–13 Hz	−0.308	−0.058	−0.250	−1.432 to +0.336	6 / 3
Beta, 13–30 Hz	−0.439	−0.272	−0.168	−0.519 to +0.133	8 / 1

Each of nine people receives equal weight, using equal-session mean trial dB from both hands. Contra. and ipsi. average the corresponding C3/C4 hand components; L is contra. minus ipsi. All 2,592 hand trials, including 246 flagged trials, are retained. Negative L means a lower contralateral than ipsilateral task/baseline change; it does not establish an absolute contralateral power decrease. Values are rounded to three decimals here and retained at full precision in the data tables. These fixed-estimator ratios are descriptive, not unbiased ERD estimates.

For the designated broad-EEGNet endpoint, descriptive associations between signed laterality and saved binary LOSO balanced accuracy were rho = −0.150 for mu and +0.600 for beta. Shared-input coefficients were −0.083 and +0.483, and CSP4+LDA coefficients were +0.067 and +0.317, respectively (Table S4; S1 Appendix, Fig S3). All six coefficients use the same nine people and have no p-values. The positive beta coefficient indicates that higher balanced accuracy accompanied a larger, less-negative signed descriptor; it does not support a benefit from stronger relative contralateral suppression.

Discussion

Training-duration Sensitivity in a Small Development Benchmark

Training duration was a major source of internal performance variation. At two epochs, S3 remained near chance with either raw or source-normalized inputs; at 16 epochs, both performed substantially better. Rank aggregation changed the source-selected schedule and recovered performance for S3 and S8. The matched-runtime control also retained a large mean difference between fixed training and the historical mean-loss schedule. An apparent cross-person model collapse should therefore be examined alongside selected duration, source-validation trajectories, per-class recall and prediction concentration. These outputs help distinguish an inadequate stopping schedule from a failure attributed prematurely to the input representation. In this benchmark, a fixed 20-epoch development condition also reached a cohort mean of 43.99%, compared with 42.67% for mean-rank and 33.72% for the earlier mean-loss condition. Those cross-batch means are descriptive. The matched-runtime evidence shows that duration auditing can diagnose apparent collapse without establishing an optimal stopping rule or a typical-person benefit.

These findings do not establish rank selection as a general solution. Within-fold ranks remove loss-scale differences but discard the size of those differences; equal fold weighting imposes a further aggregation choice. The candidate was developed after this benchmark had been inspected. In addition, the matched-runtime fixed-duration condition improved the mean while the median paired difference was negative. The external evaluations did not compare rank with mean-loss selection and cannot independently confirm the internal gain. A prospective test would fix the competing selection rules before observing new-person outcomes and match runtime, source examples and update budgets.

Complete Spectral Pipelines Have Cohort-dependent Outcomes

The frozen evaluations limit claims of a consistent shared-input advantage. Shared mu/beta input gave no internal advantage, a small uncertain positive estimate on PhysioNet, a corrected adverse contrast on Cho2017 and an uncertain near-zero contrast on Lee2019. The Cho2017 finding contradicts a consistent advantage of this fixed shared-input pipeline under the evaluated source-only adapter. It does not show that mu/beta rhythms lack information, that parameter sharing is generally harmful or that a different band definition would perform better.

The class recalls reveal a trade-off hidden by the cohort means. On Cho2017, the shared-input pipeline reduced mean left-hand recall while increasing right-hand recall; on Lee2019, opposing recall changes partly offset one another. These patterns describe the frozen predictions. They do not justify correcting labels after inspection or attributing the contrast to physiology. Montage, acquisition reference, amplitude scale, temporal alignment and band-logit averaging are possible contributors, but the study did not isolate any of them.

One shared parameter set does not imply a cheaper computation: the network processes two band views and pools their BatchNorm exposure. Independent-branch controls alter both capacity and normalization, and the broadband capacity comparator retains a small disclosed parameter mismatch. The source-fitted CSP model also transfers poorly and asymmetrically in the Cho2017/Lee2019 evaluation. Performance above this contextual comparator does not establish practical or clinical reliability. Contemporary adaptation and foundation-model methods were not compared under a common budget, so the results support no state-of-the-art claim.

The external schedules belong to the complete-pipeline contrast: broad/shared fits used 18/17 epochs for PhysioNet and 14/19 for Cho2017/Lee2019. The paired differences combine duration selection, preprocessing, one-view versus two-view computation and normalization exposure. They cannot identify the independent effect of parameter sharing or band decomposition. No matched-duration pair of final all-source checkpoints was available in the frozen evidence;

comparing a new equal-duration pair would require additional source fits if those models do not already exist.

The archived-output audit confirms a severe CSP class asymmetry, not an exact single-class prediction for every trial. Original source validation establishes artifact and partition consistency, and the external replay establishes consistency of the declared adapter and frozen predictions. Neither proves that the chosen CSP design is optimal or that every acquisition and mapping assumption is physically correct. Held-out source predictions for this exact 21-channel CSP specification were not available in the audited evidence. Transfer sensitivity is a plausible explanation, but neither an exclusive causal account nor the absence of every implementation error is established.

Person-level Evidence and Descriptive Signal Characterization

Additional seeds assess sensitivity to stochastic fitting; they do not increase the number of participants. Lee2019 sessions and source-identity subsets likewise provide repeated observations within a person. The paired person-level analysis retains all declared participants and reports the direction of each difference. Its bootstrap intervals condition on the frozen source models and omit uncertainty from another nine-person source cohort, acquisition adapter or development history. For the internal study, overlapping outer training sets further limit an independent-person interpretation.

A few large recoveries can improve a mean without producing consistent benefit across people, so tests of average change and directional consistency need not agree. The original rank-selection sign calculation retained exact ties; excluding them conventionally gives a different result from both the archived calculation and the paired t-test. The discrepancy must remain visible rather than be hidden under an incorrect test label. The later internal families retain their corrected null findings. Percentile bootstrap intervals and randomization tests also need not agree in a small sample; an interval excluding zero does not override a corrected null test. The Cho2017/Lee2019 evaluation uses a separately frozen two-cohort family: Cho2017 survives correction, whereas

Lee2019 establishes neither benefit nor equivalence. These separate tests do not establish a cohort-by-pipeline interaction. PhysioNet has a different archived primary test and an amended protocol; its p-value belongs to that evaluation rather than the newer family.

The BNCI analysis provides descriptive, sensor-level baseline-relative characterization, not the cause of the decoder contrasts. Mean signed hand laterality was negative in both bands: six of nine people had negative mu laterality and eight had negative beta laterality, with positive signs in the others. A negative relative descriptor can still accompany increased power on both sides, as in S2, so its absolute components must also be examined. The positive broad-model beta correlation means higher balanced accuracy accompanied less-negative relative laterality, rather than stronger suppression. None of the six associations supports a subgroup threshold or a model-selection rule.

Neither a lateralized scalp mean nor its correlation with LOSO balanced accuracy locates a cortical source, excludes ocular or muscle contributions, or demonstrates that EEGNet used that signal. This BNCI-only physiology analysis was specified after decoder outcomes were known. It cannot explain the adverse Cho contrast or validate an external physiological mechanism.

Headline accuracies from other studies cannot position these cohort means in a common ranking. Roots et al. report pooled-window accuracy rather than participant-level balanced accuracy; overlapping-filter and multiscale architectures use different branches, tasks and selection procedures [23,24,28]. Target fine-tuning and calibration further distinguish adaptive transfer and BEETL [26,27]. Cohort eligibility, trial window, reference, montage, training exposure and target-information access must all match before a numerical comparison can address method superiority. No published model was rerun under such a matched budget.

Limitations

Successive development conditions used the same nine BNCI people. Excluding the target from each fit does not remove outcome-informed candidate design or dependence through overlapping LOSO training sets. The internal rank-selection gain was concentrated in two people, and the external evaluations did not test that selection-rule contrast. Controls also differ in runtime,

training examples, update exposure, sensor width or approximate capacity. There is no common-budget comparison with contemporary adaptation or foundation-model methods.

The PhysioNet and Cho2017/Lee2019 stages differ in montages, windows, references, filtering scope and source checkpoints. The earlier PhysioNet evaluation's metadata amendment followed partial execution. In the frozen transfer experiment, repository freezes preceded fitting or inference but followed metadata inspection; they are not independently registered preregistration. Cross-provider participant identities remain unverified. The external bootstrap intervals condition on one frozen source cohort and omit source-sampling, adapter and development-history uncertainty.

Fig 4 reports the earlier PhysioNet evaluation separately.

Fig 4. Separate-cohort binary transfer. A: empirical cumulative distributions of three-seed person-mean balanced accuracy, with one CSP score per person; the dotted vertical line marks binary chance (50%). B: all 109 paired primary effects, with zero and the mean indicated. The interval and p-value are the frozen person-bootstrap and tie-excluding sign-test outputs. Three participants were deterministically resampled under the disclosed metadata amendment.

The external duration schedules were selected using four fixed BNCI validation groups with unequal participant counts (2/2/2/3) and equal fold weighting. The grouping was frozen before target inference, but it is one arbitrary source-validation design. Its sensitivity was not tested prospectively.

The Cho2017/Lee2019 study used a common operational tensor and a committed adapter for transfer across three providers. Matching tensor dimensions does not make the physical measurements equivalent. Native export voltage calibration, the original Cho acquisition reference and hardware cue latency were not independently verified. The Cho channel order was mapped from the publication montage because the MATLAB payload has no channel-name strings. Lee event indexing retains a one-native-sample convention discrepancy. Numerical consistency checks cannot resolve these acquisition uncertainties.

The two external stages also differ in the extent of verification. The earlier PhysioNet evaluation required a sampling-rate amendment after partial execution; its later portability check examined saved outputs without replaying inference. The Cho2017/Lee2019 study froze preprocessing before source fitting and the inference contract before external prediction, and its completed scientific validator reports independent raw-to-prediction replay. The manuscript checks recalculate saved outputs and verify hash bindings. These provide different levels of reproducibility and should not be treated as interchangeable.

Evaluation was offline and used zero-phase filtering. Online latency, prospective assistive control, clinical populations and patient benefit were not tested. The two-second Cho2017/Lee2019 transfer crop omits part of each imagery interval and differs from the providers' original benchmarks. A subsequent study could preregister a calibrated adapter and frozen source-only models for another cohort, specifying per-class and per-person outcomes before access. Retuning on the present held-out cohorts would be further development, not independent external confirmation.

The physiology estimates compare a one-segment baseline with a three-segment task estimator; unequal log-power variance can bias the dB difference. Negative signed laterality can arise from ipsilateral power increases without absolute contralateral suppression. The descriptor and decoder use different task windows, and nine descriptive person records with overlapping LOSO fits cannot establish a subgroup biomarker or mechanistic attribution. Baseline-relative ratios cancel only a constant multiplicative gain; they do not correct reference, timing, spatial mapping or artifacts. External voltage calibration remains unknown, and external physiology has not been evaluated.

Conclusions

Training-duration selection was consequential in this nine-person BNCI development benchmark, but its gains were concentrated rather than broadly shared. Complete fixed shared-input pipelines

showed no consistent external advantage. The Cho2017 shared-input pipeline had 1.513-pp lower balanced accuracy than broad EEGNet under its 19-versus-14-epoch source-selected schedule; Lee2019 remained compatible with small effects in either direction. These comparisons do not isolate spectral sharing. Archived CSP predictions document strong class asymmetry but do not identify its cause. The post-outcome BNCI sensor analysis found negative mean signed hand laterality with participant variation; it establishes neither a decoder mechanism nor external physiological replication. Acquisition uncertainty and offline evaluation limit deployment conclusions.

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Supporting information captions

S1 Appendix. Supplementary results, descriptive associations, inventories and detailed reproducible methods. Internal supplementary figures and tables retain their original S numbering.

S1 Data. Frozen participant-level summaries, saved external predictions, descriptive physiological outputs, source-data provenance and validation evidence. Files are CSV/JSON or gzip-compressed CSV; raw EEG and model weights are excluded.